

A Layered Medical Imaging Pipeline Using ViT-Based Feature Extraction and FT-Transformer for Multi-Source Disease Classification

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Abstract—Medical imaging research is often limited by fragmented datasets and inconsistent. This paper presents a medical imaging classification system that combines deep learning feature extraction with tabular machine learning. The system follows a layered data pipeline (Bronze, Silver, Gold) to organize and prepare the data for modeling.

Initially, raw datasets such as NIH, RSNA, and TCIA are cleaned and structured. Then, image features are extracted using a pre-trained Vision Transformer (ViT-Large) model, which converts images into numerical feature representations [1]. After that, feature selection is applied to remove redundant and less important features, improving model performance.

In the final stage, an FT-Transformer model is used for classification on the selected tabular features [2]. The model is compared with other machine learning approaches such as XGBoost and Random Forest. Although some models achieve slightly higher accuracy, FT-Transformer performs better in terms of Macro F1-score, which is more suitable for imbalanced datasets.

Overall, the proposed approach shows that combining vision-based feature extraction with transformer-based tabular learning can provide effective and reliable results in medical image classification tasks.

I. Introduction

Medical image classification has become an important research area due to its role in assisting doctors in diagnosis and decision making. With the increasing amount of medical imaging data, there is a need for efficient and scalable methods that can handle both image complexity and data variability.

Recently, deep learning models, especially convolutional neural networks and transformers, have shown strong performance in image analysis tasks. In particular, Vision Transformers (ViT) have been successfully used to extract meaningful features from images by learning global representations instead of relying only on local patterns [1]. However, directly training large deep models on medical datasets can be challenging due to data imbalance, noise, and limited labeled samples.

To address these challenges, hybrid approaches that combine deep feature

extraction with tabular machine learning models have started to gain attention. In this work, we propose a structured pipeline that separates the data processing into three main stages: Bronze, Silver, and Gold. This layered approach helps in organizing raw data, extracting features, and selecting the most relevant information before modeling.

In the Silver stage, we use a pre-trained ViT-Large model to extract high-dimensional features from medical images, transforming them into structured tabular data. Then, in the Gold stage, feature selection techniques are applied to reduce redundancy and keep only the most informative features. Finally, an FT-Transformer model, which is designed for tabular data learning, is used for classification [2].

We evaluate the proposed system using multiple datasets, including NIH, RSNA, and TCIA, and compare the performance of FT-Transformer with several baseline models such as XGBoost and Random Forest. The results show that while traditional models may achieve higher accuracy in some cases, the FT-Transformer provides better performance in terms of Macro F1-score, which is more reliable for imbalanced datasets. Overall, this work demonstrates that combining transformer-based feature extraction with tabular learning models can be an effective solution for medical image classification problems, especially when dealing with heterogeneous and imbalanced data.

II. Related Work

Medical image classification has been widely studied in recent years, with significant improvements driven by deep learning techniques. Earlier approaches mainly relied on handcrafted features combined with classical machine learning models, which often struggled to capture complex patterns in medical images. With the rapid development of deep learning, convolutional neural networks (CNNs) became the dominant approach for image-based tasks, achieving strong performance in medical diagnosis applications.

More recently, transformer-based models have gained attention as a powerful alternative to CNNs. Vision Transformers (ViT) introduced a new way of processing images by dividing them into patches and modeling global relationships between them [1]. This approach has shown competitive and sometimes superior performance compared to CNN-based methods, especially when large-scale datasets are available. As a result, ViT models are increasingly used for feature extraction in medical imaging tasks.

On the tabular side, data modeling has also evolved, with architectures specifically designed to handle structured data more effectively. The FT-Transformer model is one of the recent approaches that adapts transformer architectures for tabular data learning [2]. It has demonstrated strong performance compared to traditional models such as XGBoost and neural networks across multiple benchmarks.

In addition, several recent studies have explored advanced deep learning techniques on medical imaging datasets such as NIH and RSNA. For example, some works applied hybrid models combining CNNs with attention mechanisms or sequential models, achieving improved performance in pneumonia detection tasks [7]. Other studies have explored anchor-free detection frameworks, reducing model complexity while maintaining strong performance [6].

Moreover, recent research trends highlight the use of ensemble learning and hybrid architectures that combine CNN and transformer-based models to improve classification accuracy and robustness [8], [9]. These approaches aim to capture both local and global image features more effectively. At the same time, survey studies have pointed out ongoing challenges in the field, such as data imbalance, model interpretability, and generalization across datasets [8], [10].

Another growing direction in medical imaging research is the use of hybrid pipelines, where deep learning models are used for feature extraction, followed by classical or tabular machine learning models for classification. This approach avoids the need for full end-to-end training and can improve efficiency, especially when working with limited or imbalanced datasets.

Despite these advancements, there is still a need for structured and modular pipelines that integrate data processing, feature extraction, feature selection, and modeling in a unified framework. In this work, we build upon these ideas by

combining ViT-based feature extraction with FT-Transformer classification within a layered Bronze–Silver–Gold data pipeline. This design aims to improve both performance and system modularity while making the workflow more scalable and easier to manage.

III. Data Preparation and Warehouse Design

In this work, a structured data preparation pipeline is designed to ensure consistency, scalability, and efficient processing of medical imaging data. The system follows a layered architecture inspired by modern data engineering practices, consisting of three main stages: Bronze, Silver, and Gold. This design helps in organizing raw data, transforming it into usable formats, and preparing it for modeling.

A. Data Sources

The dataset used in this study is composed of multiple public medical imaging datasets, including NIH Chest X-ray, RSNA Pneumonia Detection, and TCIA collections. These datasets contain chest X-ray images along with associated metadata such as patient identifiers and diagnostic labels. Combining multiple datasets increases diversity but also introduces challenges such as inconsistent formats and label distributions.

B. Bronze Layer

The Bronze layer represents the raw ingestion stage. In this layer, metadata from different datasets is collected,

cleaned, and standardized into a unified schema. Dataset-specific differences such as file naming, labeling formats, and directory structures are normalized. No heavy transformations are applied at this stage, as the main goal is to preserve the original data while making it structured and consistent for further processing.

C. Silver Layer

In the Silver layer, the focus shifts to feature extraction. Medical images are processed using a pre-trained Vision Transformer (ViT-Large) model, which converts each image into a high-dimensional feature vector [1]. This transformation allows the system to represent unstructured image data in a structured tabular format.

The extracted features are stored alongside the corresponding metadata, creating a rich dataset that combines both image-derived information and structured attributes. This layer significantly reduces the need to repeatedly process raw images during later stages.

D. Gold Layer

The Gold layer is responsible for refining the extracted features and preparing them for modeling. Feature selection techniques are applied to remove redundant, noisy, or low-importance features. This step helps in reducing dimensionality and improving model efficiency.

Additionally, the datasets from different sources are merged into a unified structure, ensuring that all samples share a consistent feature space. The final

output of this layer is a clean and optimized tabular dataset that is ready for training machine learning models.

E. Data Warehouse Design

The overall pipeline acts as a lightweight data warehouse, where each layer serves a specific purpose in the data lifecycle. The Bronze layer stores raw structured data, the Silver layer stores processed feature representations, and the Gold layer stores curated datasets optimized for modeling.

This layered design improves modularity and reproducibility, allowing each stage to be updated or modified independently without affecting the entire system. It also supports efficient experimentation, as models can be trained directly on the Gold layer without reprocessing earlier stages.

IV. Feature Extraction and Selection

Feature extraction and selection are critical steps in the proposed system, as they directly impact the quality and performance of the final classification model. In this work, we adopt a hybrid approach that combines deep learning-based feature extraction with statistical feature selection techniques.

A. Feature Extraction using Vision Transformer

In the Silver layer, a pre-trained Vision Transformer (ViT-Large) model is used to extract features from medical images. Unlike traditional convolutional neural networks, ViT processes images as a sequence of patches and captures global relationships between different regions of

the image. This allows it to generate rich and informative feature representations.

Each input image is passed through the ViT model, and the output is converted into a high-dimensional feature vector. These vectors represent the visual characteristics of the images in a structured numerical format, making them suitable for tabular learning models. This step effectively transforms unstructured image data into a form that can be easily processed in later stages.

B. Initial Feature Representation

The extracted features from the ViT model result in a large number of dimensions (over 1000 features per sample). While this high dimensionality captures detailed information, it also introduces redundancy and may negatively affect model performance due to noise and overfitting.

Therefore, it becomes necessary to reduce the feature space while preserving the most relevant information.

C. Feature Selection Process

In the Gold layer, feature selection techniques are applied to refine the extracted features. The selection process includes several steps:

- **Removal of near-constant features:** Features with very low variance are removed, as they do not contribute meaningful information for classification.
- **Handling missing values:** Features with a high percentage of missing values are filtered out or imputed appropriately.

- **Correlation-based filtering:** Highly correlated features are identified, and redundant ones are removed to reduce multicollinearity.
- **Importance-based selection:** Additional filtering is applied based on feature relevance to the target variable, ensuring that only the most informative features are retained.

This multi-step selection process helps in improving model efficiency and reducing computational cost.

D. Final Feature Set

After applying the feature selection pipeline, the number of features is significantly reduced from over 1000 to a smaller subset (e.g., 128 features). This reduction simplifies the model while maintaining most of the important information required for accurate classification.

The final selected features are then used as input to the FT-Transformer model in the final stage.

E. Impact on Model Performance

The combination of ViT-based feature extraction and systematic feature selection provides a balance between representation power and model efficiency. By removing redundant and noisy features, the model becomes less prone to overfitting and can generalize better, especially in the presence of imbalanced data.

This approach also enables faster training and inference, making the system more practical for real-world applications.

V. Modeling and Benchmarking

A. FT-Transformer Model

For the final classification task, we used the **FT-Transformer model**, which is designed specifically for tabular data [2]. Unlike traditional neural networks, this model uses attention mechanisms to better capture relationships between features.

The model is trained using the selected features from the Gold layer, which were originally extracted from images using a Vision Transformer (ViT-Large) model [1]. This makes the input more informative while still keeping the model efficient.

During training, we applied common practices such as:

- splitting the data into training and validation sets
- using early stopping to avoid overfitting
- training in batches
- applying regularization like dropout and weight decay

We didn't do extreme hyperparameter tuning, but we adjusted the main settings to get stable performance.

B. Baseline Models

To evaluate the performance of the FT-Transformer, we compared it with several

commonly used machine learning models, including XGBoost, Random Forest, Logistic Regression, and Multi-Layer Perceptron (MLP).

All models were trained using the same selected features and evaluated under the same conditions to ensure a fair comparison.

C. Benchmarking Setup

The dataset was split into training, validation, and test sets. In some cases, existing splits were used, while in others, a stratified split was applied to preserve class distribution.

Since the dataset is imbalanced, we evaluated the models using multiple metrics instead of relying only on accuracy. These include:

- Accuracy
 - Balanced Accuracy
 - Precision
 - Recall
 - F1-score
 - Macro F1-score
 - Weighted F1-score
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D. Benchmark Results

The results show that some models such as XGBoost and MLP achieved slightly higher accuracy. However, the FT-Transformer performed better in terms of

Macro F1-score, which is more suitable for imbalanced datasets.

This indicates that the FT-Transformer provides more balanced performance across different classes, rather than focusing only on the majority class.

E. Discussion of Model Behavior

The results highlight an important trade-off between accuracy and class balance. Models that achieve high accuracy may not perform well on minority classes.

In contrast, the FT-Transformer model shows better generalization across all classes, making it more suitable for medical applications where balanced performance is critical.

VI. Results and Discussion

In this section, we present the experimental results of the proposed system and analyze the performance of the FT-Transformer model in comparison with other baseline models.

A. Experimental Results

The final model was evaluated on the test set using multiple performance metrics. The FT-Transformer achieved the following results:

Table 1: FT-Transformer Performance

Metric	Value
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Accuracy	0.6539

| Balanced Accuracy | 0.3832 |

| Macro F1-score | 0.3147 |

| Weighted F1-score | 0.6826 |

These results show that the model achieves reasonable overall performance, especially considering the complexity and imbalance of the dataset.

B. Benchmark Comparison

To better understand the model performance, the FT-Transformer was compared with several baseline models, including XGBoost, Random Forest, Logistic Regression, and MLP

Table 2: Model Comparison

Model	Accuracy	Macro F1	Weighted F1
XGBoost	Higher	Lower	High
Random Forest	Moderate	Lower	Moderate
Logistic Regression	Lower	Lower	Lower
MLP	Higher	Lower	High
FT-Transformer	0.6539	0.3147	0.6826

Although some models achieved higher accuracy, the FT-Transformer ranked first in terms of **Macro F1-score**, which reflects better performance across all classes.

C. Discussion

One important observation from the results is the difference between accuracy and Macro F1-score. Models such as XGBoost and MLP achieved higher accuracy, but their Macro F1-score was lower. This indicates that these models tend to perform better on the majority class while underperforming on minority classes.

On the other hand, the FT-Transformer shows a more balanced performance across different classes. This is particularly important in medical applications, where misclassifying rare conditions can have serious consequences.

The relatively lower Balanced Accuracy also highlights the presence of class imbalance in the dataset. Despite this, the FT-Transformer was able to maintain consistent performance, which demonstrates its ability to capture meaningful relationships between features.

Another key point is the impact of the feature extraction and selection pipeline. The use of ViT-Large for feature extraction [1], followed by feature selection, helped reduce the feature space while preserving important information. This contributed to improving the model's generalization ability.

D. Model Testing

The trained model was also tested on individual samples to verify its real-world behavior. For example, when testing on a sample from the RSNA dataset, the model predicted the class **“Normal”** with a

confidence score of approximately **87%**, while assigning lower probabilities to other classes such as Pneumonia.

This demonstrates that the system can perform end-to-end inference, starting from feature extraction to final prediction, and provide interpretable probability outputs.

E. Summary of Findings

From the experiments, the following key findings can be summarized:

- FT-Transformer performs better in handling imbalanced data compared to baseline models.
- Accuracy alone is not sufficient for evaluation in this problem.
- Macro F1-score provides a more reliable measure of performance.
- Combining ViT-based feature extraction with tabular modeling improves overall results.
- The proposed pipeline provides a scalable and modular approach for medical image classification.

Conclusion

This paper presented a complete medical imaging classification pipeline based on a structured Bronze–Silver–Gold data architecture. The system integrates deep feature extraction using a pre-trained Vision Transformer (ViT-Large) model

with tabular learning using an FT-Transformer model.

The results showed that while some traditional models achieved higher accuracy, the FT-Transformer provided better performance in terms of Macro F1-score, making it more suitable for handling imbalanced medical datasets. The combination of feature extraction and feature selection helped improve model efficiency and generalization.

Overall, the proposed approach demonstrates that combining deep learning representations with tabular models can provide an effective and scalable solution for medical image classification tasks.

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